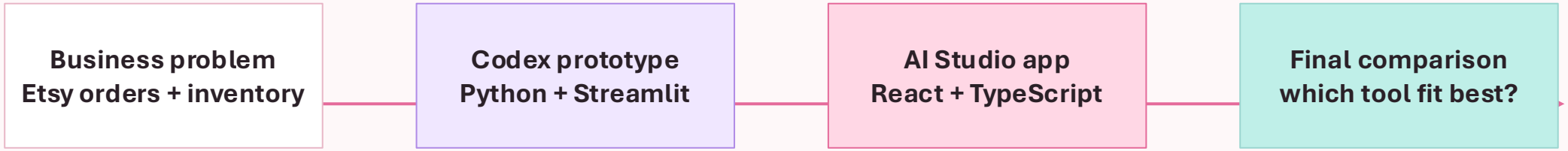
AI Productivity Application

Compared Codex + Streamlit vs Google AI Studio for app development

Final Applied AI Project

Hunter Bloom



Project question

Which paid GenAI development tool created the most useful business app?

Problem definition and relevance

The business problem

NailsByHuntrr has data scattered across multiple platforms; including product photos, color swatches, nail sets, keychain ideas, revenue records, reviews, and printing information.

Why AI fits

Generative AI can rapidly prototype software, suggest structure, debug code, and transform a dashboard idea into a more polished app.

Overview	About me	Nails	Keychain	Gel Polisj	Filament Colors
Revenue	Orders	Reviews	Pinterest	Instagram	API

Goal: one app to help a business owner stay organized and using data/records to make decisions.

Live demo plan

Main strategy

Streamlit prototype - Quick Summary
Google AI Studio app – Main Demo

Live Google app demo link: https://huntrr-dashboard-607549261991.us-east1.run.app/

Demo stop 1	**Streamlit prototype**	45–60 sec	Show that the Python dashboard was the first working direction. Open it, show the home/table area, then move on quickly.
Demo stop 2	**Google app overview**	60 sec	Open the deployed React app and show the main navigation so the class sees the final app feel.
Demo stop 3	**Main Google app demo**	2–4 min	Use the live deployed app here: Dashboard → Orders → Colors → Revenue/Analytics → Bambu Printer API → Bambu printer/status area.
Demo stop 4	**AI workflow evidence**	30 sec	Mention proof: live deployed app, Codex session ID 019f3f7a-9d49-78c0-9b58-3a435faa736b, and AI Studio App ID 0742ceed-95b8-4e70-8f60-072837437992.

Original plan: Codex + Streamlit prototype

First implementation path: AI-assisted Python data dashboard.

Codex session ID
019f3f7a-9d49-78c0-9b58-3a435faa736b

I used Codex to help create the first version as a Streamlit dashboard. This session ID documents the Python prototype work, including generated code, Streamlit structure, data tables, charts, and debugging.

Python / Streamlit direction

```
import streamlit as st
import pandas as pd

st.title("NailsByHuntrr Dashboard")
orders = pd.read_csv("orders.csv")
st.dataframe(orders)
```

Best fit

Data cleaning, quick dashboard sections, pandas analysis, and debugging Python errors.

Main limitation

The Streamlit version felt more like a dashboard than a polished business app.

DEMO 1 **Quick Streamlit pass: open it, show one data/table view, then return to slides.**

Final pivot: Google AI Studio app version

Second implementation path: app-style interface generated through AI Studio.

Google AI Studio App ID
0742ceed-95b8-4e70-8f60-072837437992

Live deployed app: https://huntrr-dashboard-607549261991.us-east1.run.app/

The final app is a React + TypeScript project built from the Google AI Studio export. It uses Vite, Tailwind, Recharts, Motion, Lucide icons, Express, and the Google GenAI package.

package.json evidence

```
"dependencies": {
  "react": "^19.0.1",
  "typescript": "~5.8.2",
  "vite": "^6.2.3",
  "@google/genai": "^2.4.0",
  "recharts": "^3.9.2"
}
```

Scan live app

DEMO 2 Open the deployed Google app here and show navigation/home before the feature slide.

Dashboard	Orders	Colors	Revenue
Reviews	Gallery	Analytics	Bambu Printer

My preference: Google AI Studio made the final project feel more like a real app.

Code comparison: Python dashboard vs TypeScript app

This was the biggest technical difference between the two project versions.

Streamlit prototype

Language	Python
Framework	Streamlit
Strength	Data analysis + charts
App style	Dashboard / prototype
Best use	pandas, tables, CSVs

Google AI Studio final app

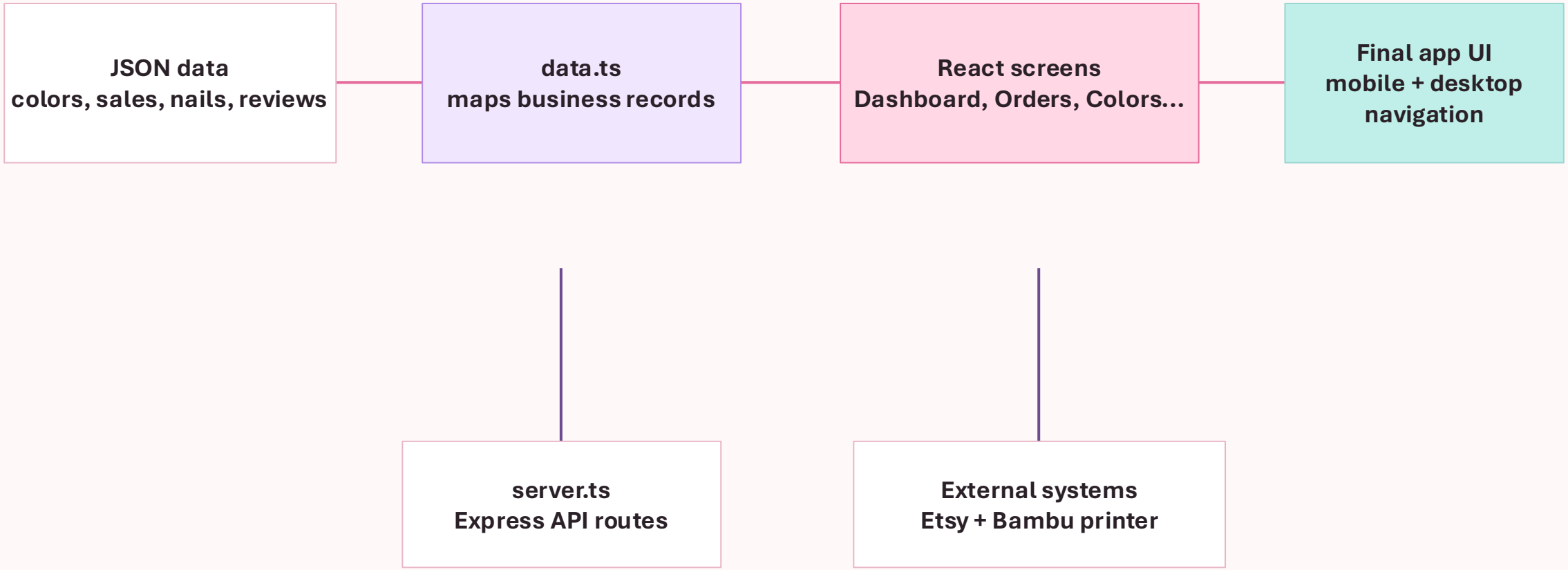
Language	TypeScript / JavaScript
Framework	React + Vite
Strength	Interactive app experience
App style	Mobile-friendly UI
Best use	navigation, screens, polished demo

Key takeaway

Python helped with business data; JavaScript/TypeScript made the final version feel like a real application.

Google app architecture from the uploaded source

The exported project separates data, components, API routes, and app navigation.



Evidence in source files

App.tsx handles screen navigation; data.ts maps JSON into app models; useEtsy.ts and server.ts prepare live integrations.

Working Bambu API / printer integration

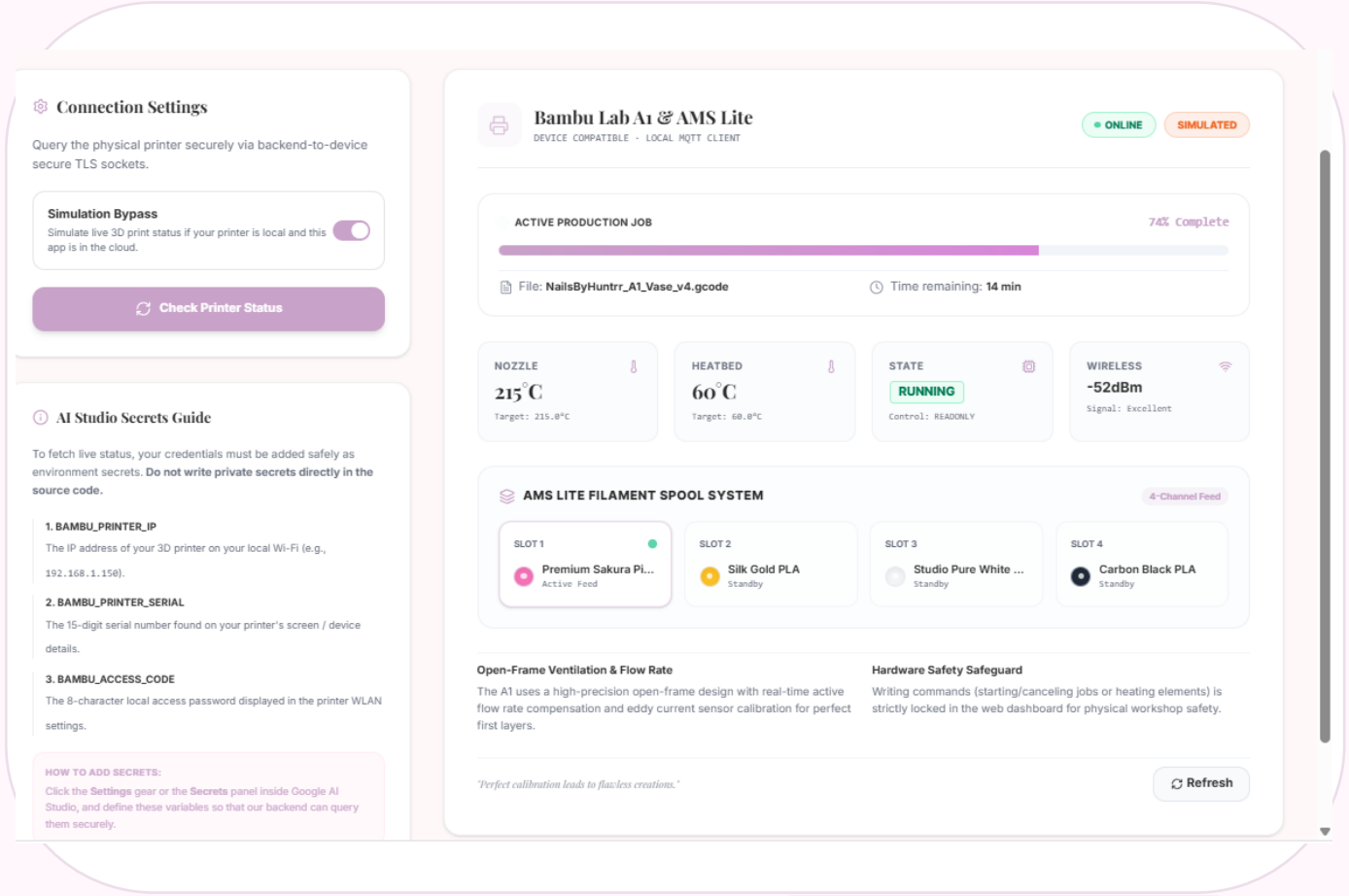
This shows the app is not only a design mockup: it includes a production-status feature for the 3D printer workflow.

What the app is reading

- Online / simulated status
- Active print job progress and time remaining
- Nozzle, heatbed, wireless, and AMS Lite spool state

Safety + secrets design

- Credentials are kept as environment secrets, not written in source code
- Physical printer commands are read-only/locked in the web dashboard
- Simulation bypass lets the class see the workflow even when local access is limited



LIVE APP SCREENSHOT

Data / Final Stats

Source-derived metrics from the latest uploaded Google AI Studio project files and deployed app source.

107	26	20	17	$1,187	32
color swatches colors.json	nail listings bundled_nails.json	sales rows bundled_sales.json	reviews bundled_reviews.json	all-time revenue shop_stats.json	all-time orders shop_stats.json

Business insights enabled

Inventory risk — **3 nail products at / below reorder level**

Customer proof — **4.94/5 average rating from review data**

Demand view — **20 tracked historical transactions**

Shop traffic — **5,038 visits and 93 favorites**

Latest app source used for these stats: huntrr-dashboard (1).zip • Deployed app: https://huntrr-dashboard-607549261991.us-east1.run.app/

Evaluation method: 10-prompt GenAI Benchmark Comparison

Rubric focus: manually score prompt results and compare AI tools.

Scoring scale

1 = not usable | 3 = usable with edits | 5 = strong, demo-ready result

Prompts tested

1. Homepage	Codex 4 / AI Studio 5
2. Order tracking	Codex 5 / AI Studio 4
3. Brand styling	Codex 3 / AI Studio 5
4. Inventory colors	Codex 4 / AI Studio 4
5. Sales charts	Codex 5 / AI Studio 4
6. Mobile feel	Codex 3 / AI Studio 5
7. Data organization	Codex 4 / AI Studio 5
8. Debugging errors	Codex 5 / AI Studio 3
9. Future AI ideas	Codex 4 / AI Studio 5
10. Final demo app	Codex 3 / AI Studio 5

Prompt scoring results

AI Studio scored higher overall because it created a more polished final app experience.

Average prompt score



Winner count across 10 prompts

6	3	1
AI Studio wins	Codex wins	Tie

Finding: Codex was best for debugging and Python-style data work. AI Studio was best for visual design, mobile feel, and final demo readiness.

AI Speed and Workflow Comparison

benchmark the benefit of the AI tools, including workflow speed.

Speed metric used

Manual workflow speed score based on prompt-to-usable-output, wait time, number of retries, and how quickly I could turn AI output into a working app feature.

AI setup	Tool	Speed	What happened
Codex paid session	OpenAI Codex	**4/5**	Strong for Python code and debugging; slower when the Streamlit app needed setup fixes.
AI Studio free / Gemini Flash 3.5	Google AI Studio	**2/5**	Initial workflow felt too slow for a final app build, so it was not ideal for the quick deadline.
AI Studio Pro / latest Pro model	Google AI Studio	**5/5**	After upgrading to Pro and switching from Gemini Flash 3.5 to the latest Pro model, the app-building workflow felt much faster and more useful.

Upgrade reflection

I started with the free Google AI Studio workflow using Gemini Flash 3.5, but it felt too slow for a project due the next day. Upgrading to Pro and switching to the latest Pro model made the final app version feel faster, cleaner, and more presentation-ready.

Cost comparison: paid AI tool decision

measurable evaluation, resource choice, and reflection on why the final tool changed.

ChatGPT Plus + Codex

$20/month

Best value for Python + Streamlit coding, debugging, and restructuring the first dashboard prototype.

Google AI Studio Free + Gemini Flash 3.5

$0 start

Good for testing, but the free workflow felt too slow for a final project app build.

Google AI Pro + latest Pro model

$4.99/mo intro → $19.99/mo

Fastest final workflow; better interface quality, stronger demo readiness, and the version I preferred.

What the price difference showed

Lower cost / free access	Useful for starting, but slower responses made the final build harder under a deadline.
ChatGPT Plus at $20	Worth it for code help because Codex could explain, debug, and improve the Python project.
Google AI Pro after upgrade	Best final value because speed and app design improved enough to become the version I demo.

Final cost reflection: I started with a slower/free Google workflow, then upgraded because the Pro model produced a faster and more polished final app. ChatGPT/Codex was still stronger for coding support, but Google AI Studio Pro gave the best final app experience.

Pricing references: OpenAI ChatGPT pricing; Google One AI plans

Development workflow: Codex vs AI Studio

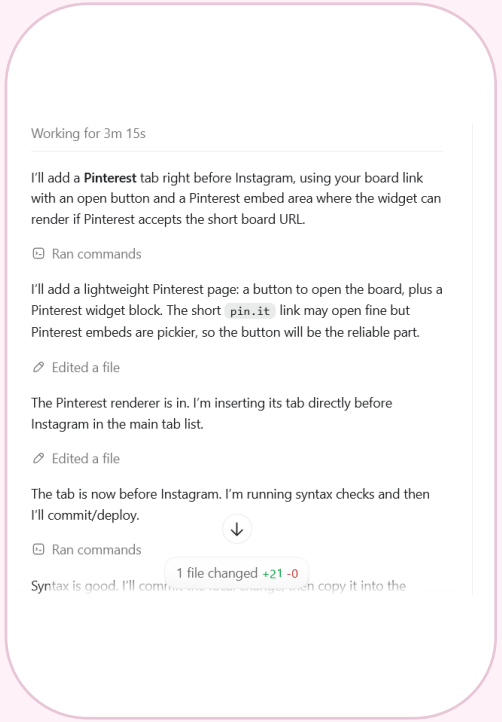
Codex workflow

Why I liked it

Shows what it is doing while it works: commands, files edited, syntax checks, and deploy steps.

Made it easier to trust the process because I could follow the reasoning and progress.

Different from AI Studio because Codex was changing files directly in my project workspace on my computer.



Evidence: Codex showed progress while editing files.

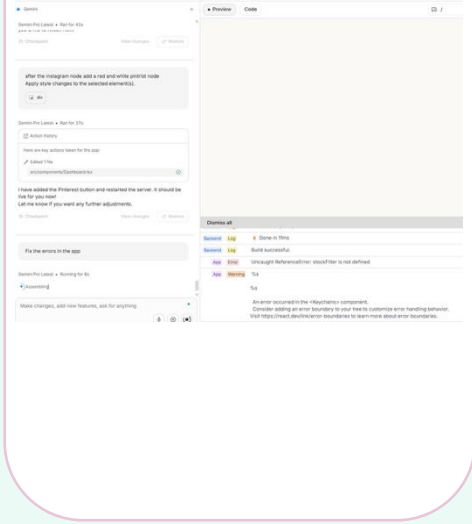
Google AI Studio workflow

Why I still preferred it

Built a more polished app-style experience with preview, code view, and action history.

Stronger for visual layout, tabs, mobile feel, and quick UI experiments.

It changed the app files inside the AI Studio build environment, then I exported/deployed the project.



Evidence: AI Studio showed edited files and preview errors.

Codex felt better for transparent coding work; AI Studio felt better for turning the idea into the final app experience I wanted to demo.

Error handling: AI still needed debugging loops

Some prompts finished successfully, but the app could still produce a runtime error that needed another AI repair prompt.

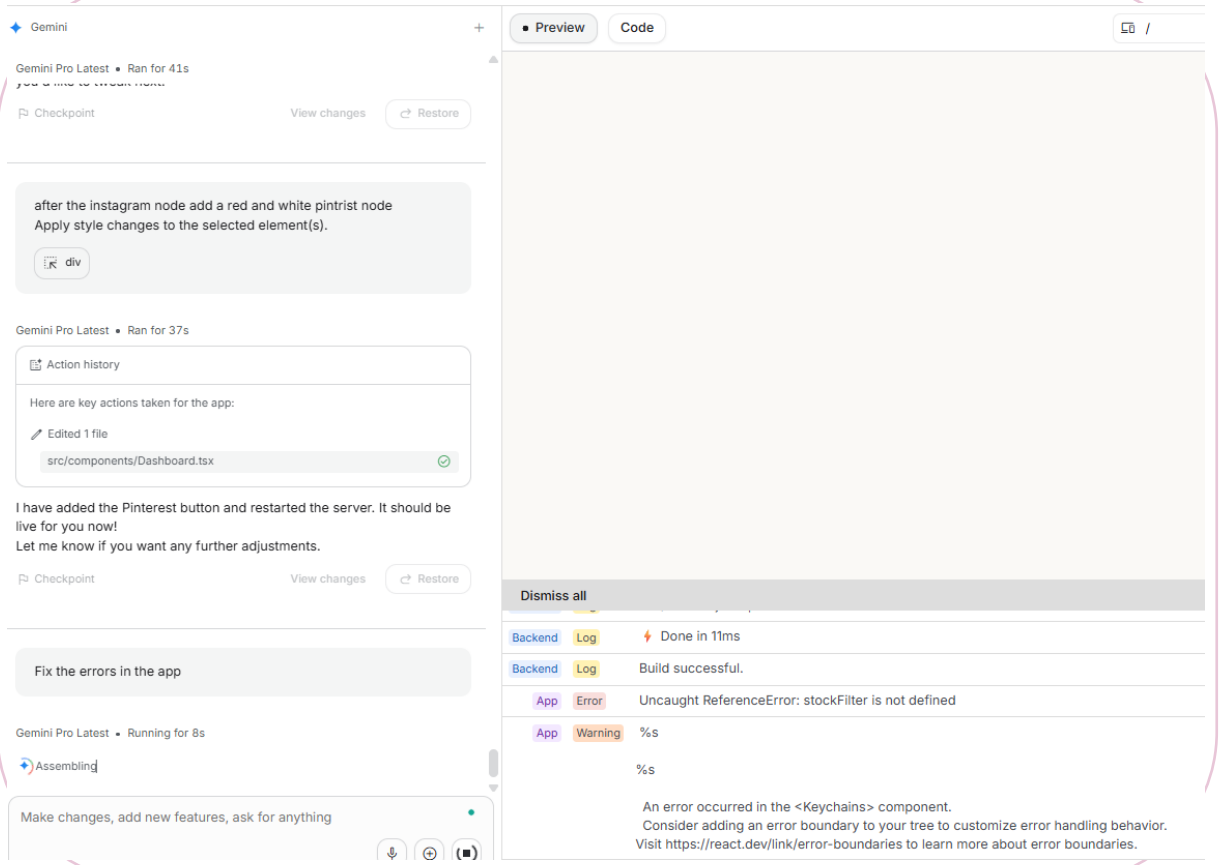
What happened

After a feature request, AI Studio reported that the build was successful.

The preview then showed a runtime error: a missing variable in the Keychains component.

I had to ask the AI agent to go back into the code and fix the error.

Bug-fix prompts usually took longer than normal feature requests because the AI had to inspect the code, change the file, and retest.



Screenshot evidence: build completed, then the preview showed an app error that required another fix prompt.



Benefits after applying AI tools

Rubric focus: measurable benefit and discussion of findings.

Before AI-assisted development	After AI-assisted development
Scattered data sources	Single app workspace
Manual tracking	Search/filter/order tools
Hard to demo on mobile	Mobile drawer + bottom nav
Separate product + revenue views	Product, color, revenue, reviews, analytics

Measurable result

The final app organized 8+ core business areas and mapped real project data from sales, reviews, nail listings, colors, and shop statistics.

What each AI tool did best

Discussion of findings: no single AI tool was best at everything.

Codex + Streamlit

- Python code generation
- Debugging errors
- CSV / pandas workflow
- Revenue charts

Google AI Studio

- Visual design
- Navigation structure
- Mobile app feeling
- Presentation-ready demo

Final decision

I preferred AI Studio for the final app, but Codex was still valuable for technical building and debugging.

Challenges and solutions

Late pivot	Framed the switch as an AI tool comparison instead of hiding it.
Streamlit setup	Kept it as a working Python prototype path, then moved final demo to AI Studio.
Etsy API approval	Documented integration as a future production feature.
Bambu printer connection	Verified with the working Bambu status screen: online/simulated status, print progress, temps, Wi-Fi, and AMS spool slots; secrets stay out of the source code.
Rubric comparison	Used a 10-prompt scorecard and noted open-source baseline as a limitation.
AI Studio runtime errors	Used the agent to return to the code, define missing variables, retest, and continue building.

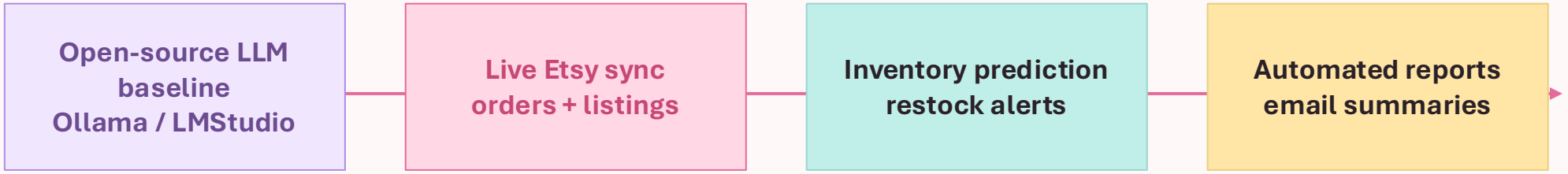
Quality, privacy, and limitations

Responsible AI use matters because this app handles business data.

Privacy	Customer/order data should not be shared publicly.
Secrets	API keys belong in environment variables, not code.
Accuracy	AI-generated code must be tested before real business use.
Manual scores	1–5 ratings are useful but subjective.
APIs	Etsy and Bambu features require approval, keys, and local access.
Open-source baseline	Not completed in final build; should be added for stronger rubric alignment.

Responsible conclusion: AI helped me build faster, but I still had to inspect the code, test features, and document what was not finished.

Future improvements



Machine learning extension: use past sales, inventory levels, color tags, and review data to predict restock risk and recommend which products to make next.

Conclusion

AI-assisted development helped turn a business problem into a working app direction.

Final finding

Codex was strongest for building and debugging the Python/Streamlit prototype. Google AI Studio was strongest for making the final app feel polished, visual, and easier to present.

What I learned

Different GenAI tools are better at different stages of development: ideation, coding, debugging, interface design, and final presentation.

Applied AI impact

The project connects AI, data, coding, and entrepreneurship by improving how a real small business can manage production and decisions.

FINAL PEEK

If time remains, end by returning to the Google app dashboard for 15 seconds.

References and project evidence

documentation, sources, and proof of work.

OpenAI Codex documentation	https://developers.openai.com/codex
Google AI Studio build mode	https://ai.google.dev/gemini-api/docs/aistudio-build-mode
Google AI Studio quickstart	https://ai.google.dev/gemini-api/docs/ai-studio-quickstart
Streamlit documentation	https://docs.streamlit.io/
React documentation	https://react.dev/learn/build-a-react-app-from-scratch
Vite documentation	https://vite.dev/guide/
Uploaded Google project source	huntrr-dashboard (1).zip
Codex Session ID	019f3f7a-9d49-78c0-9b58-3a435faa736b

Google AI Studio App Link https://ai.studio/apps/0742ceed-95b8-4e70-8f60-072837437992

Live deployed app: https://huntrr-dashboard-607549261991.us-east1.run.app/ | File naming: Bloom_Hunter_NailsByHuntrrAIProject_Slides

Live app QR